



程序设计原理

函数

任课教师：喻 梅
于瑞国
王建荣
李雪威
赵满坤



Introduction

Why Do We Need Functions?

- Avoid code duplication
- Make programs modular
- Improve readability and maintenance
- Break complex problems into smaller tasks



Introduction

Suppose that you need to find the sum of integers from 1 to 10, 10 to 20, from 20 to 37 ,and from 35 to 49.

```
for(int i=1;i<=10;i++)
    sum +=i;
cout << "The sum is:"<<sum<<endl;
for(int i=20;i<=37;i++)
    sum +=i;
cout << "The sum is:"<<sum<<endl;
for(int i=35;i<=49;i++)
    sum +=i;
cout << "The sum is:"<<sum<<endl;
```

```
int sum(int i1,int i2)
{
    int sum = 0;
    for(int i=i1;i<=i2;i++)
        sum += i;
    return sum;
}

int main()
{
    cout<<"The sum is:"<<sum(1,10)<<endl;
    cout<<"The sum is:"<<sum(20,37)<<endl;
    cout<<"The sum is:"<<sum(35,49)<<endl;
    return 0;
}
```



Defining a Functions

What Is a Function?

A function is a block of code that performs a specific task.

A function has four parts:

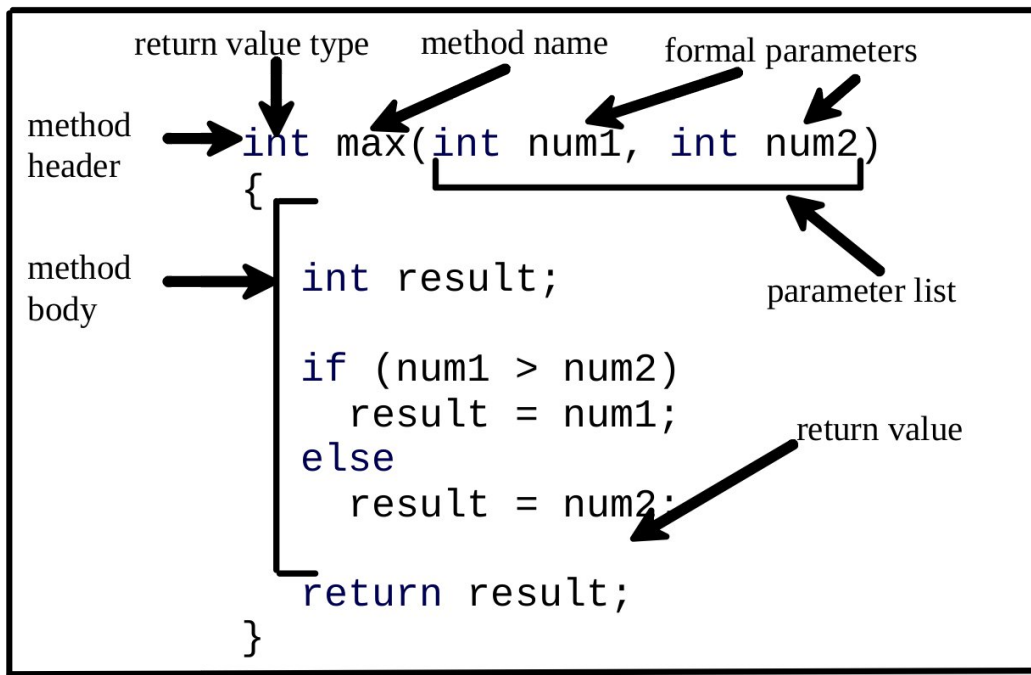
1. Return value type
2. Function name
3. Parameters
4. Function body



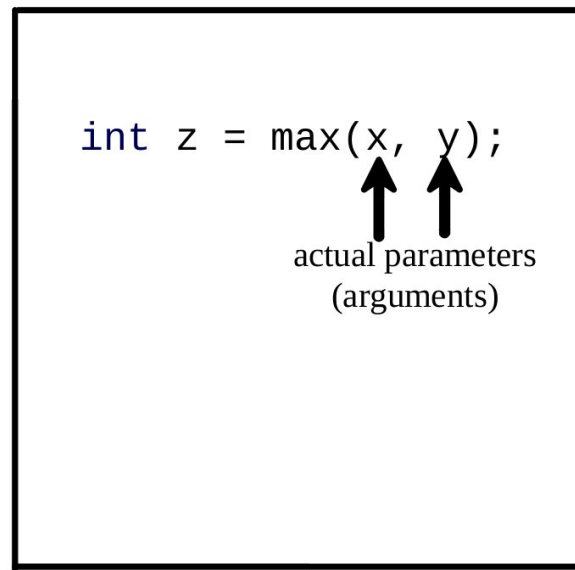
Defining a Functions

```
returnValueType functionName(list of parameters)
{
    //function body;
}
```

Define a function



Invoke a function





Introducing Functions, cont.

- ***Function signature*** is the combination of the function name and the parameter list.
- The variables defined in the function header are known as ***formal parameters***.
- When a function is invoked, you pass a value to the parameter. This value is referred to as ***actual parameter or argument***.

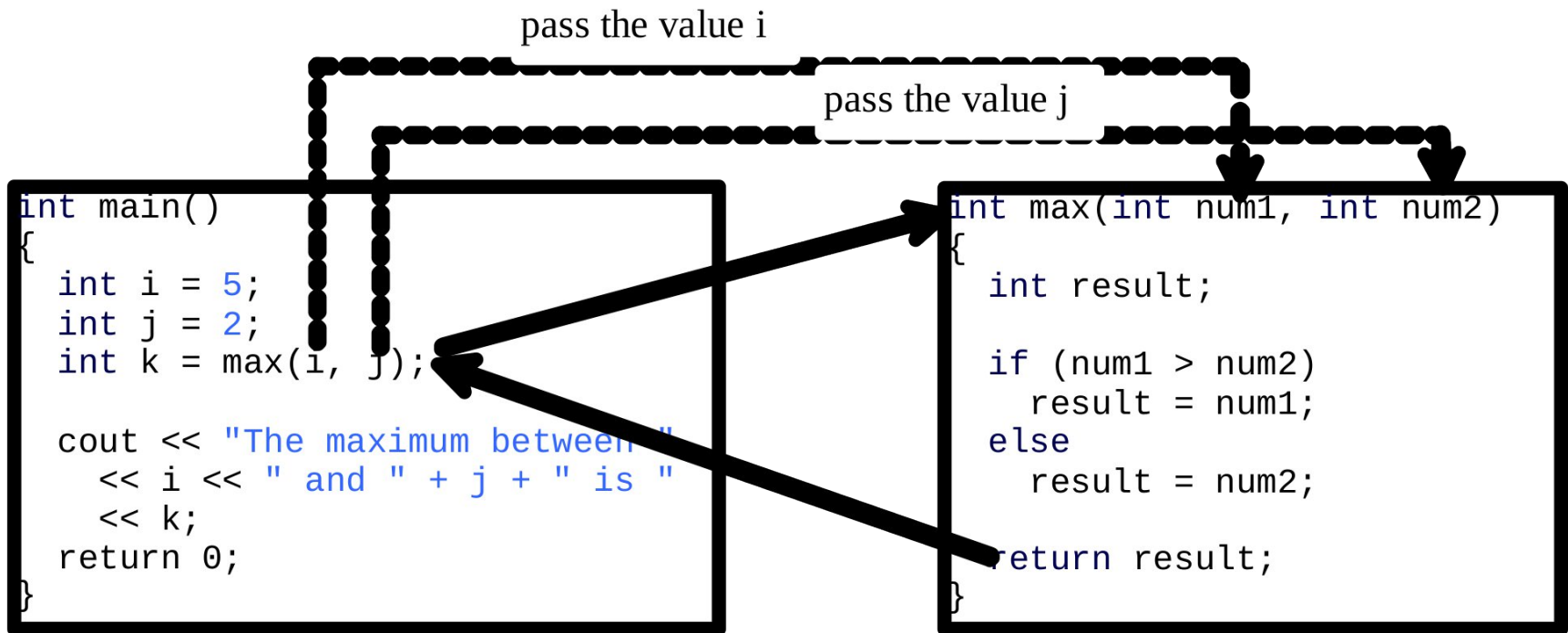


Introducing Functions, cont.

- A Function may return a value. The returnValueType is the data type of the value the function returns. If the function does not return a value, the returnValueType is the keyword void. For example, the returnValueType in the main function is void.



Calling Functions, cont.

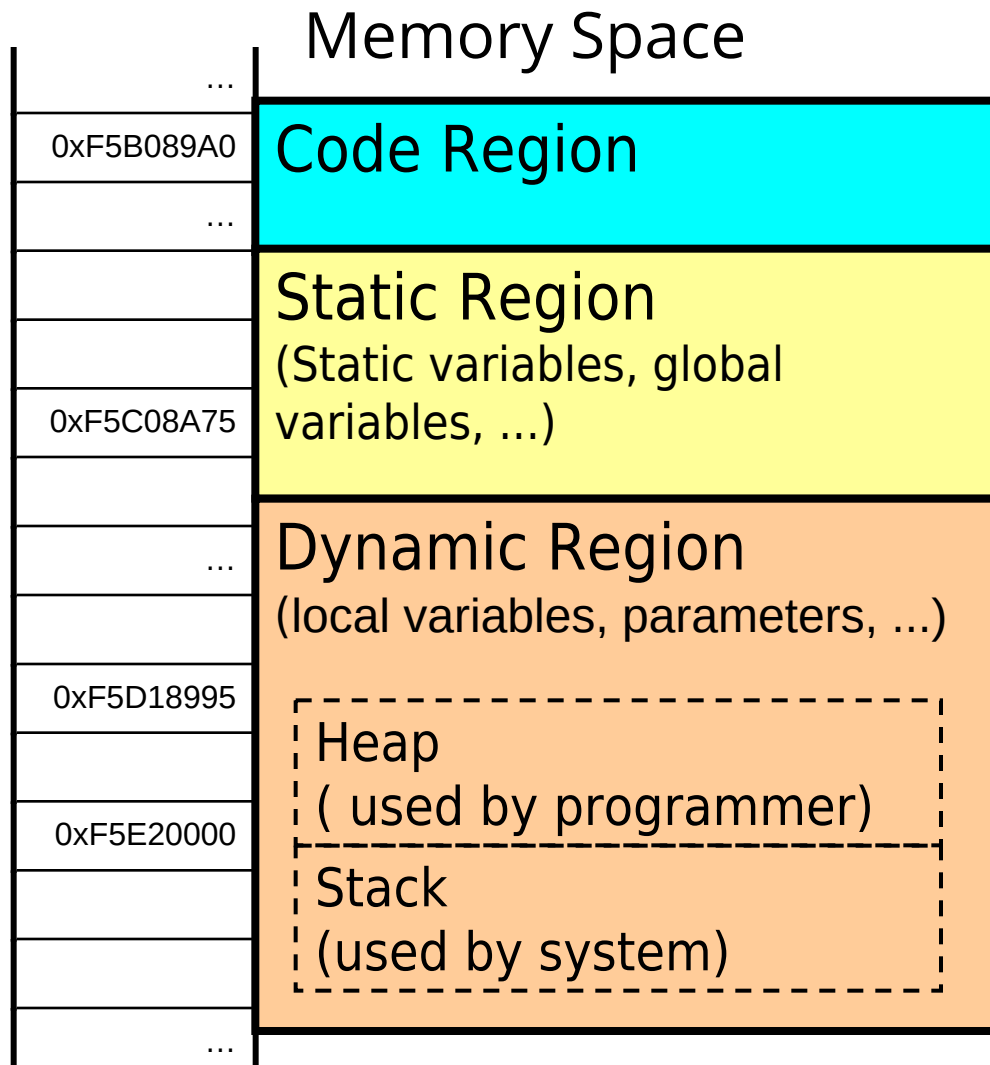




Memory

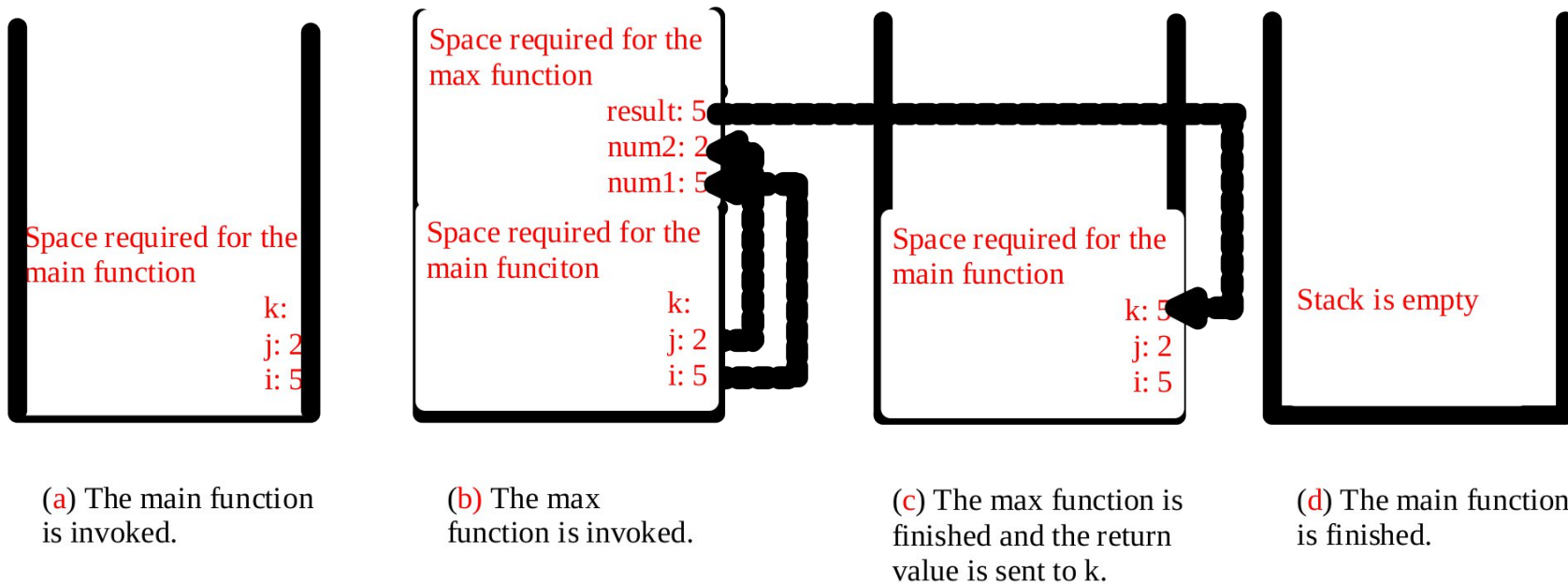
Memory space

- 1 A “continuous” space
- 1 Three regions





Call Stacks





Passing Arguments by Value

When you invoke a function with a parameter, the value of the argument is passed to the parameter. This is referred to as *pass-by-value*. If the argument is a variable rather than a literal value, the value of the variable is passed to the parameter. The variable is not affected, regardless of the changes made to the parameter inside the function. We will examine an interesting scenario in the following example, in which the parameters are changed in the function but the arguments are not affected.



Passing Arguments by Value

```
#include <iostream>
using namespace std;

/** Swap two variables */
void swap(int n1, int n2)
{
    cout << "\tInside the swap function" << endl;
    cout << "\t\tBefore swapping n1 is " << n1 <<
        " n2 is " << n2 << endl;

    // Swap n1 with n2
    int temp = n1;
    n1 = n2;
    n2 = temp;

    cout << "\t\tAfter swapping n1 is " << n1 <<
        " n2 is " << n2 << endl;
}
```

```
int main()
{
    // Declare and initialize variables
    int num1 = 1;
    int num2 = 2;

    cout << "Before invoking the swap function, num1 is "
        << num1 << " and num2 is " << num2 << endl;

    // Invoke the swap function to attempt to swap two
    variables
    swap(num1, num2);

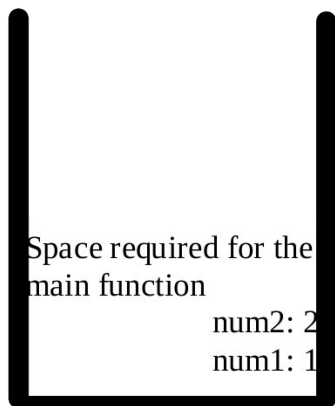
    cout << "After invoking the swap function, num1 is " <<
        num1 <<
        " and num2 is " << num2 << endl;

    return 0;
}
```

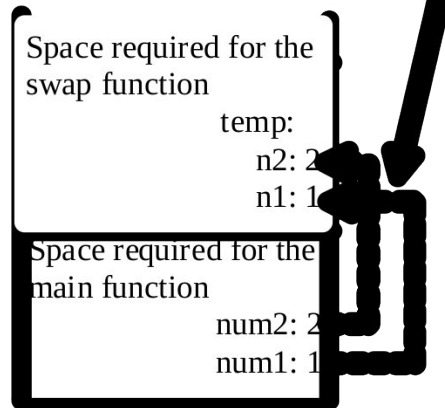


Passing Argument by Value, cont.

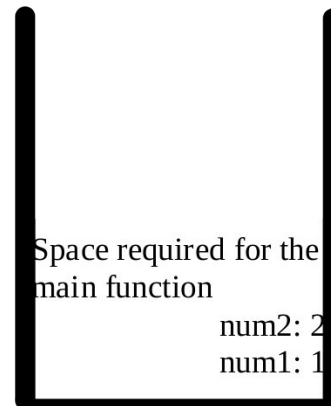
The values of num1 and num2 are passed to n1 and n2. Executing swap does not affect num1 and num2.



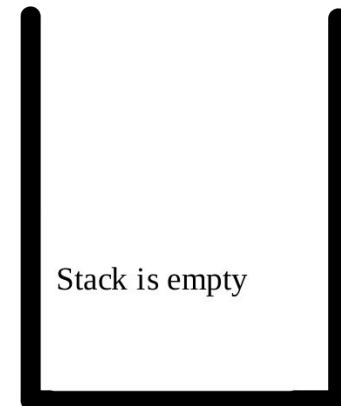
The main function is invoked



The swap function is invoked



The swap function is finished



The main function is finished



Exercise

题目描述

给你两个数，求最大公约数。要求用函数实现。

输入

输入数据包含多个测试实例，每组数据占一行，每行有两个正整数 n 和 m 。 n 和 m 同时为0标示输入数据的结束，本行不做处理。

输出

对于每组输入样例，输出一个数，是他们的最大公约数

样例输入

```
15 5
20 3
65 13
0 0
```

样例输出

```
5
1
13
```

提示

$1 \leq n, m \leq 10^{15}$



Exercise

题目描述

编写一个求 x 的 n 次方的函数。题目中会给出两个数 x 和 n ，输出 x 的 n 次方即可。

输入

输入数据包含多个测试实例，每组数据占一行，每行有两个正整数 x 和 n 。 x 和 n 同时为-1标示输入数据的结束，本行不做处理。保证 $0 \leq x, n \leq 15$

输出

对于每组输入样例，输出一个数，是 x 的 n 次方的计算结果。

样例输入 Copy

```
2 3
3 5
5 7
5 0
-1 -1
```

样例输出 Copy

```
8
243
78125
1
```

提示

$0 \leq x, n \leq 15$



Exercise

题目描述

寻找并输出11~999(闭区间)之间的数 m ，它满足 m 、 m 的平方和 m 的立方均为回文数。

– 回文：各位数字左右对称的整数。

例如：11满足上述条件

$$11*11=121, 11*11*11=1331.$$

用函数实现

输入

无

输出

输出所有的11~999之间的数 m ， m 的含义参照题目描述，每个数占一行，按照从大到小的顺序输出。